

1. Directory structure of unix operating system.
2. write not on grep command
3. Booting sequence of init process.
4. Features of unix operating system.
5. function of unix kernel.
6. ls command.
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9. Features of shell
10. sed command
11. soft link and hard link
12. i-node block
13. write a command :- awk, bc, uniq, kill, find, ps
14. Different types of file permission also explain how they can be change
15. Booting sequence
16. Explain tr with options and example

1. Directory structure of unix operating system.

→ In unix storing the data and application in file.

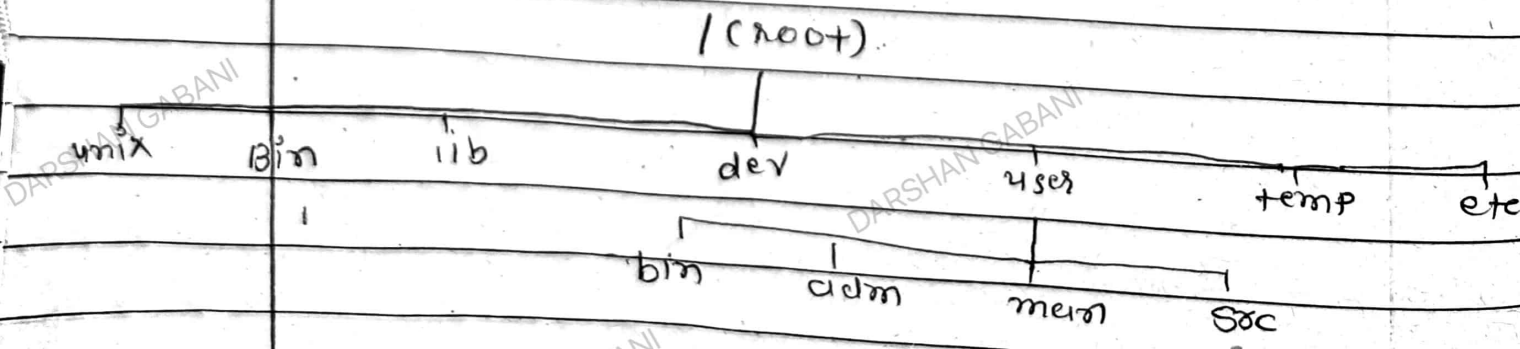
→ All the files are grouped together in the directory structure

→ The file system is arranged in a hierarchical structure, like an inverted tree.

→ The top is traditionally called root

→ All the data of unix is organized into file

→ All the files are organized into directories.



* (/root) directory

→ The root directory of top of unix file system

→ And other are local.

/bin

→ It is a directory which contain all normal user executable command.

→ Binary file are executable file

* /dev directory (Device file)

→ contain device file that represent hardware device like terminal and printer

* /tmp (Temporary file)

→ temporary file created by unix or user

→ This file automatically delete than system will shut down.

/lib

→ contains shared library file and sometimes other kernel-related file

/kernel

→ contain kernel files.

2. write note on grep command.

→ grep command to a searched file for a pattern and display matching and non-matching lines.

Syntax:-

grep options pattern filename(s)

Ex:-

\$ grep "suresh" student.lst

1111	Amit B. Shah	01/02/1980	Suresh	bca	98242
1114	sumil K. Modi	08/2/2022	suresh	bca	88246
\$					

* If the pattern doesn't found it simply return the prompt.

\$ grep "Arund" student.lst

\$ -

grep command also used with multiple file name, It display file with output

```
$ grep "Ahemdabud" student.1st student2.1st
```

student.1st : 1115	Vimal	18/03/2023	Ahemdabud bba
student1.1st : 2235	Jay	12/12/2012	Ahemdabud bba
student.1st : 2365	Mital	08/06/2024	Ahemdabud bba
student2.1st : 2654	Deep	18/06/2024	Ahemdabud bba

* grep Options :-

-i (ignoring case)

→ when you want to search a name but don't know case, use the -i options.

→ This options ignores case for pattern matching.

Ex :-

```
$ grep -i 'NAVSARI' student.1st
```

2111	Bimal. suzati	10/08/2023	<u>Navsari</u>	bba
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→ (Deleting a file)

→ when a file to a delete to use -v.

```
$ grep -v "Bharuch" student.1st > student2.1st
```

\$

-n (display line number of Record)

```
$ grep -n "Ahemdabud" student.1st
```

```
5: 1115 | Vimal K. Patel | 18/03/2023 | Ahemdabud | bcom
```

```
12: 3124 | Jay K. Rana | 29/09/2010 | Ahemdabud | bcom
```

\$

* -c (Counting)

→ -c to counting a lines of number pattern

```
$ grep -c "Ahemdabud" student.1st
```

2

\$

-c also use with multiple file name

```
$ grep -c "Ahemdabud" student.1st student2.1st
```

→ student.1st : 2

student2.1st : 2

\$

-e (Matching multiple pattern)

→ -e options matching multiple pattern.

\$ grep -e "makwana" -e "makvana" -e "makhwana"
student.lst

makwana
1737 | Sunil. ~~chavla~~ | 11/03/2022 | Vapi | Bce

1232 | Mayur. makvana | 12/01/2020 | Bhavnagar | B.com

1837 | Jay. Makhwana | 18/10/2012 | Surat | Bba
\$

3. Booting sequence of 1011

→ The init process in unix-based system is the first user-space process started by the kernel during boot.

2. ROM diagnostics are run :-

7) Hardware and memory tests are performed

2. Boot loader is loaded.

→ The boot loader is loaded into RAM by the ROM.

3. kernel is local.

→ The kernel program is called by the user

→ Recall that the user needs to enter the name of the program.

1.	kernel initialization takes place
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> it involves performing memory tests, initializing the device through device driver

5 The init program

- m) This process has the PID 1 (Process identification)
- m) which is the second process of the system.

6 The getty process

- The init program also invokes the getty program which establishes a communication with the terminals of the unix system.

7 The login program

- once the user types login name and password, getty transfers control to a login program to verify the login-name and password entered by user.
- To login successfully to the \$ prompt appears on the screen.

init → getty → login → shell

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4. Features of Unix operating system.

UNIX offers several features like

1. multiuser
2. Multitasking
3. Inter process communication
4. security
5. Portability
6. open source
7. Multitasking

1. Multiuser

- The same computer resources say hard disk, memory etc are access to many user
- The user are given different terminal to operate from.
- A terminal is keyboard and monitor, which are the input and output device for the user
- All terminal are connected to main computer

- Dumb terminal

- The dumb terminal consist a keyboard and a display.
- They have to be not connect with server machine

- Terminal Emulation

- m) A PC as own microprocessor, memory and disk drive
- m) Attaching a PC to host via cable.
- m) The software that makes a PC work like dumb terminal called terminal emulation

- Dial-in Terminal

- This terminal is use telephone lines to connect with host machine.

2} Multitasking

- UNIX, capable of carrying out more than one job at same time
- A single user can also run multiple task concurrently.
- It is use for edit a file, send email to a friend.

- In such multitasking situation only one job runs in the foreground while the rest run in the background.
- You can switch the jobs between background and foreground.
- Multitasking is important component of the process management system.

3. Inter process communication

- The feature allows user to pass on data, exchange mail or program to another user in network.

4. Security

- UNIX allow sharing of data but to secure them provide three levels of protection.

1. login name and password
2. read, write and executed permission to each file
3. Succeeds in opening, it your secure are safe

5. Portability

- unix is written in high level language
- that can be easy to read, changed and move to other machines

6. open system

- UNIX has an open architecture using which user can add tools to program

5- Functions of unix kernel

- Process Management
- Memory Management
- File system Management
- Device Management
- Security and access control
- Networking
- system call interface

6 characteristic of unix

- multitasking
- multiuser
- portability
- security and permission
- shell and commandline interface
- networking
- toolbox philosophy

• **ls** command

- m) All data unit organized into files.
- m) All files are organized into directories.
- m) These directories are organized into a tree-like structure ~~is~~ called the filesystem.
- m) you can use **ls** command to list out all the file or directories available in directory.
- m) **ls** commonly most use command for viewing file

Syntax :-

ls [options] [directory]

*** Basic use ***

1. ls

- m) List the name of file and directories in the current directory.

Ex :-

\$ ls

Output :-

File1.txt File2.txt dir1 dir2

\$

15 - 1

Provide detail and information like file permission number etc

Ex:-
\$ ls -l

output :-

permission	owner file	associated with File	size of the file	date and time
-rw-r--r--	user	group	1234	sep 23
12:34	file.txt	File of directory name		

3 \$ ls -la (list all file including hidden file)

show all file including hidden file

\$ ls -u

output :-

hidden file	file.txt	dir
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4.8 \$ ls -l

1056406 document 3828450 sum.p1
→ Display inode number of file

8. vi- editor

→ vi-editor is a powerful widely used text editor in unix system.

Getting start with vi

1. opening vi

vi file name

→ to create a file

2. Modes in vi

1. Normal Mode :- Default mode for navigating and manipulating text.

2. insert mode :- for inserting text. press i enter the mode.

3. command mode :- for saving and quitting

3. Basic vi command

- switching modes

i :- insert mode at the cursor position.

a :- insert mode after the cursor

ESC :- go back normal mode from insert mode

- saving and exiting

W :- save the file

q :- quit the file

ZZ :- save and quit

q! :- quit without saving

- navigation

gg :- move the beginning of the file

G :- move the end of the file

0 :- move the beginning of the line

\$:- move the end of the line

search

/pattern :- search forward for "pattern".

?pattern :- search backward for "pattern".

exiting vi

• save and exit : wq and ZZ
Exit without saving : q!

9. Features of shell.

1. Interactive environment :-

m) The shell allow user to create communication between user and Host UNIX system.

2. Shell script

m) The shell script is group of UNIX commands and executes as individual programming file.

3. I/O Redirection

m) Redirection is function of the shell that redirect the output of the program as an input to other file or command.

4. Piping mechanism

m) piping facility allows the output of one command to be use input to another command.

e: who | wc

who output
command | wc

5. Background process

- m) multitasking facilities allows the user to run commands in background.
- m) when a background task is completed, the user is notified.

6. Shell Variable

- m) The user can declare the variables to store the data which can be access anywhere in program.

sed:-
 i → insert
 a → append
 c → change
 p → print line of structured output
 q → quit



10 sed command

m sed is tool that design for multipurpose
 m sed command used to perform find and replace, insert, delete and another operation without opening the file in a text editor

* Line addressing +

Quit (q)

m The q is quit after display specified lines.

For example 6 q display first 6 line and quit

Ex:-

5 sed '5q' student.1st

1111	Amit. Shah	01/03/1980	Surat	80%
1112	Ramesh. Bishnoi	20/10/2020	Burda	70%
1113	Rathod. Jai	11/03/1987	Vapi	60%
1114	Mam. Dorela	12/02/1980	Surat	55%
1115	Ram. Tauria	18/03/1984	Bhavan	45%

Point CP)

P command is used in sed to display line

to display line 2 to 3 from student.1st

\$ sed -n '2,3p' student.1st

1111	Amit. shah	02/02/1980	surat	bca	900+
1112	Ramesh. Bish	20/02/1982	Baroda	bba	800
1113	Rathod. Jay	11/03/1981	vapi	bca	800

to display last line of student.1st

\$ sed -n '5p' student.1st

3124 | Ritu. Makwana | 19/09/1978 | surat | bca | 952432

select multiple group line

m> Display line 2 to 3, 5 to 8 and last line from student.lst

```
$ sed -n '2,3p 5,8p $P' student.lst
```

m> 2 to 3 and 5, 6, 7, 8 and last line and.

* using multiple instruction (-e)

The -e option allow you to enter multiple instruction in one command.

```
$ sed -n -e '2,3p' -e '5,7p' -e '9,20p' student.lst
```

will output line 2, 2, 3, 5, 6, 7, 9, 20 and.

A Deleting a file

syntax

```
sed ' / pattern / d' filename
```

EX :-

```
sed ' / pattern / d' student.lst
```

* insert

Syntax :-

sed 'pattern / i / text' file

Ex :-

sed ' / pattern / ^{new} i / text' student.1st

* Append

Syntax

sed ' / pattern / a / text' file

Ex :-

sed ' / pattern / a / new text' file name

11. Soft link and hard link

m) Hard link and soft link are two different types of reference that can be created for file in unix OS.

* Hard link *

m) Hardlink can have multiple filenames to refer to the same inode

Properties :-

1. Same inode number :- Hardlink share the same inode number as the original file

2. File content :- Hardlink share the same content.

* Soft link *

m) Soft link are shortcuts or pointers to the original file's path name.

Properties :-

Different inode numbers :-

Soft link have different inode number compared to the original file

File content

→ Soft link don't contain file's data.

Ex :-

* Hard link

echo "Hello, world!" > file2.txt

In file2.txt file2.txt

* soft link

In -s file2.txt file3.txt

checking

ls -li file2.txt file2.txt ← Hard link

ls -li file3.txt ← soft link

12. i-node block or physical file structure of unix

- m) i-node block are used to describe the file characteristic and the location of the data blocks which store the contents of the system.
- m) The information related to all these file is stored in an i-node table on the disk

Detail :-

- o File type (regular, directory, device etc)
- o Number of links
- o The number of UID of the owner
- o The number of GID of the owner
- o File mode
- o size of the file
- o Date and time to last modification of file data
- o Date and time to last access to file data
- o Date and time to last change of i-node

Other detail in chattr-GP4 :-

physical

→ Boot block

→ Super block

explain chmod command using relative and absolute method.

→ The chmod command is used in UNIX to change the permission of file or directory.

→ There are two main methods

i) Relative method

ii) Absolute method

i) Relative method

Symbolic Representation

u: user

g: group

o: other

A: All

Operation

+: Add permission

-: Remove permission

=: set the exact permission

Ex:-

Add :- chmod u+x filename

Remove chmod g-w filename

Read and write :- chmod o=rw filename

Absolute method

The Absolute method uses numeric values to set permission explicitly.

Read (r) : 4

write (w) : 2

execute (x) : 1

Breakdown

Read write and execute

$$7 \text{ user} : 4(\text{read}) + 2(\text{write}) + 1(\text{execute}) = 7$$

$$5(\text{group}) : 4(\text{read}) + 1(\text{execute}) = 5$$

$$5 \text{ other} : 4(\text{read}) + 1(\text{execute}) = 5$$

2. set read and write permission user and read-only for group

chmod 644 filename

$$6(\text{user}) : 4 \text{ read} + 2 \text{ write} = 6$$

$$4(\text{group}) : 4 \text{ read} = 4$$

$$4 \text{ other} : 4 \text{ read} = 4$$

14. Different types of file permission also explain how they can be changed.

m In UNIX contain three types of file permission:-

i) Read(r) :- Allow viewing content the file

ii) write(w) :- Allow modifying and deleting the file

iii) Execute(x) :- Allow running the file as a program.

Permission level

owner(u) :- user who create a file

group(g) :- permission on file

other(o) :- Everyone else

How can be change:-

m To changed in chmod command.

m you can use either symbolic method or numeric method.

i) Symbolic method

Add permission:- `chmod u+x filename`
→ add execute permission to the user

Remove permission:- `chmod g-w filename`
→ remove write permission in group

ii) Numeric method

→ numeric method are represent by number

$$\text{Read (r)} = 4$$

$$\text{write (w)} = 2$$

$$\text{Execute (x)} = 1$$

Each set of permission is represent by a three-digit number

$$\text{Read, write and execute} = 7 \quad (4+2+1)$$

$$\text{Read, write} = 6 \quad (4+2)$$

$$\text{Read and execute} = 5 \quad (4+1)$$

$$\text{Read-only} = 4$$

* write permission to group
`chmod g+w filename`

* Remove execute permission
`chmod o-x filename`